EMS611U: Conceptual Design of Aircraft

Basic Aerodynamic Principles

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Aerodynamics Fundamentals & Road Map

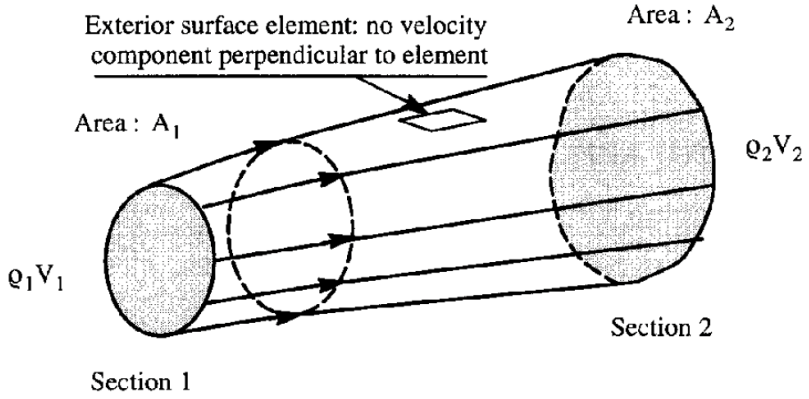
Basic Aerodynamic Principles

- Three conservation laws underpin everything.

1. Conservation of Mass:

Physical Principle:

Mass can be neither created nor destroyed.



$$\dot{m} = \rho_1 A_1 V_1 = \rho_2 A_2 V_2 = constan$$

$$\frac{kg}{m^3}\ m^2\ \frac{m}{s} \longrightarrow \left[\frac{kg}{s}\right]$$

$$\rho = \frac{m}{V}$$

Assumptions we can make here:

- $\rho = constant$ — *Incompressible Flow*

$$M = \frac{V}{a} < 0.3$$

$$\rho = \frac{p}{RT}\ ; ideal\ gas$$

$$A_1 V_1 = A_2 V_2 = constant$$

Basic Aerodynamic Principles

- Three conservation laws underpin everything.

2. Conservation of Momentum:

Physical Principle:

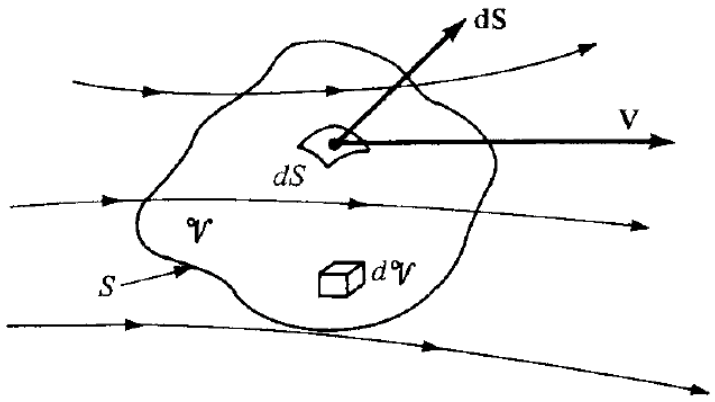
Force = time rate of change of momentum

$$F = ma \longrightarrow F = \frac{d}{dt}(mV) \longrightarrow$$

$$\frac{\partial}{\partial t}\iiint_{\mathcal{V}} \rho \mathbf{V}\, d\mathcal{V} + \oiint_S (\rho \mathbf{V} \cdot \mathbf{dS})\mathbf{V} = -\oiint_S p\, \mathbf{dS} + \iiint_{\mathcal{V}} \rho \mathbf{f}\, d\mathcal{V} + \mathbf{F}_{viscous}$$

Navier-Stokes Equations

$[N] \quad kg\,\frac{m^2}{s}$



Again, we can make assumptions and simplify

- Incompressible Flow, $\rho = constant$
- Steady Flow
- No body forces
- Inviscid Flow $\mu = constant$

Viscosity is what makes flows "stick" to surfaces and to each other.

$$\oiint_S (\rho \mathbf{V} \cdot \mathbf{dS})\mathbf{V} = -\oiint_S p\, \mathbf{dS}$$

Common Assumptions in Aerodynamics

- Incompressible
- Inviscid

⟷ Compressible Flow

$\rho \neq constant$ $\quad M = \frac{V}{a} > 0.3$

- All real substances are compressible to some greater or lesser extent; that is, when you squeeze or press on them, their density will change. This is particularly true of gases, much less so for liquids, and virtually unnoticeable for solids. The amount by which a substance can be compressed is given by a specific property of the substance called the compressibility,
- Can not assume that the heat transfer is not important anymore.
- Shockwaves might occur.

Common Assumptions in Aerodynamics

- Incompressible
- Inviscid

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 Viscous Flow

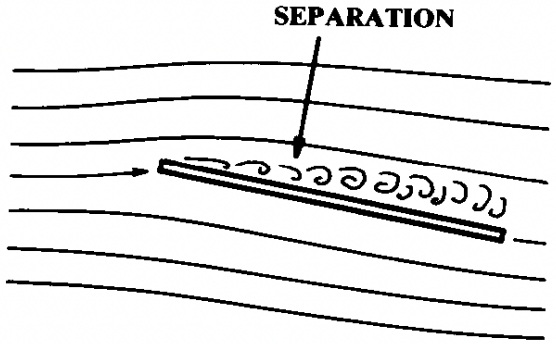
Without Viscosity

 No separation

 No boundary layer

 No turbulence

 No viscous diffusion



- All real flows are viscous!
- **D'Alembert Paradox**
- With the exception of induced drag and supersonic wave drag, which can be obtained from inviscid theory, the calculation of all other forms of drag must explicitly take into account the presence of viscosity,

Road Map

Conservation Equations
- Conservation of Mass
- Conservation of Momentum

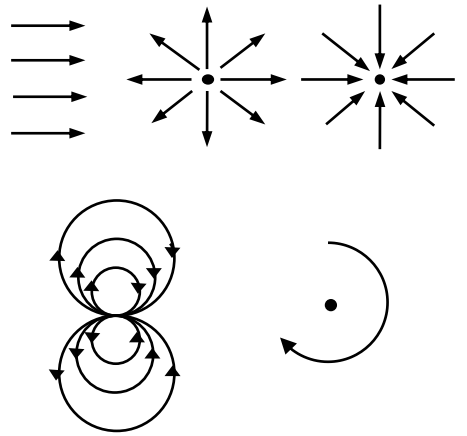
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Assumptions
- Steady
- Incompressible, Inviscid

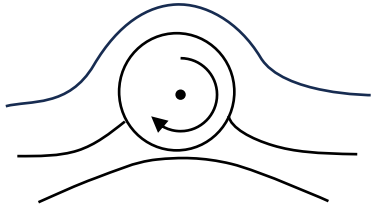
Bernoulli's Equation
- $P_1 + \frac{1}{2}\rho V_1^2 = constant$
- $V \rightarrow P \rightarrow$ Forces!

Stream function (ψ) and Velocity Potential (ϕ)
- $\nabla^2\psi = 0$; $\nabla^2\phi = 0$
- Laplace's equation, solutions are superimposable

Elementary Flows



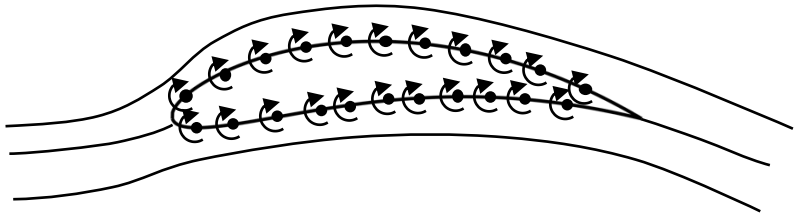
Basic Flow Construction



- Kutta-Joukowski Theorem

$$L' = \rho V_\infty \Gamma$$

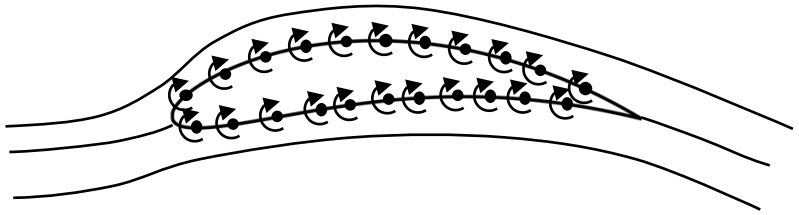
Aerodynamic Consideration of an Aerofoil



Make the aerofoil surface a streamline of the flow.
- $\gamma \rightarrow \Gamma \rightarrow L'$
- **Need a computer to solve!**

Road Map

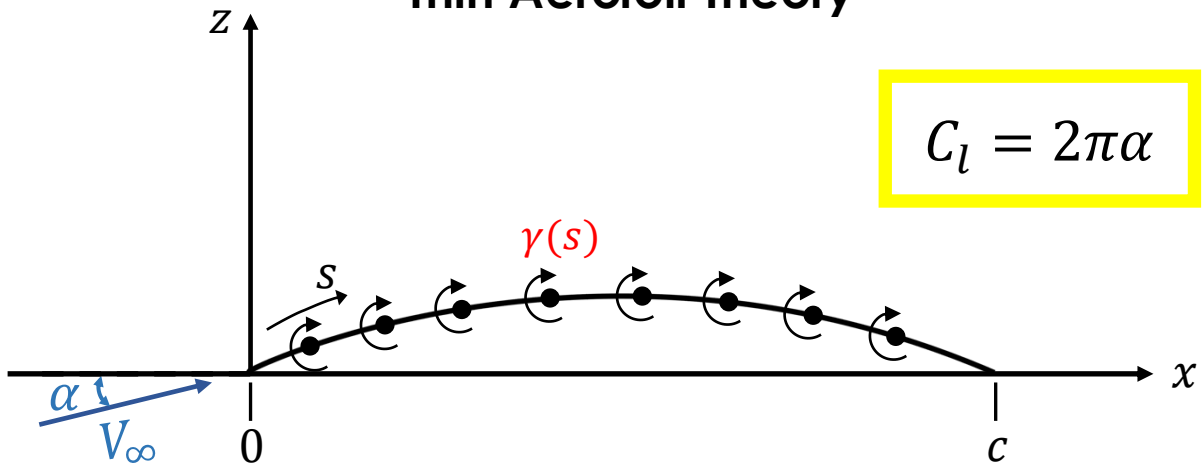
Aerodynamic Consideration of an Aerofoil



Make the aerofoil surface a streamline of the flow.

- $\gamma \rightarrow \Gamma \rightarrow L'$
- **Need a computer to solve!**

Thin Aerofoil Theory



$$C_l = 2\pi\alpha$$

Assumptions

- Steady Flow
- Incompressible Flow
- Inviscid Flow